As Long As People Play

Game Phase Classification Methodology

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1 Introduction

This document outlines the logical framework and algorithmic methodology used to classify game levels into distinct phases (Onboarding, Early Game, Mid Game, and Late Game) based on user retention heuristics. The analysis is restricted to the first 250 levels of each levelset.

2 Methodology

2.1 Retention Calculation

For a given levelset S, let U_i denote the total number of users who reach level i. The initial user count for the levelset is U_0 (the user count at the lowest level in S). The retention percentage R_i for any level $i \in S$ is defined as:

$$R_i = \left(\frac{U_i}{U_0} \right) \times 100 \tag{1}$$

2.2 Phase Classification Logic

The classification is applied sequentially to ensure mutual exclusivity among phases. Let L be the set of levels evaluated, bounded such that $l \leq 250$ for all $l \in L$. The maximum level within the evaluated set is l_{max}.

1. **Onboarding Phase:** Designed to capture the initial tutorial or highly engaging entry levels. A level l is classified as Onboarding if it is among the first three levels or if its retention remains exceptionally high:
$$l \leq 3 \quad \vee \quad R_l \geq 85\% \tag{2}$$

2. **Early Game Phase:** Defined as the period following onboarding but preceding the first major drop in player retention. Let l_{drop} be the first level where retention falls below 60%. A level l is classified as Early Game if it is not Onboarding, and occurs before this drop:
$$l \notin \text{Onboarding} \quad \wedge \quad l < l_{drop} \tag{3}$$

3. **Mid Game Phase:** Represents the core gameplay loop where retention stabilizes before the deep late-game churn. A level l is classified as Mid Game if it has not been previously classified, and either maintains moderate retention or falls within the first 30% of the levelset's progression:
$$l \notin \{\text{Onboarding}, \text{Early Game}\} \quad \wedge \quad (R_l \geq 30\% \quad \vee \quad l \leq 0.3 \times l_{max}) \tag{4}$$

4. **Late Game Phase:** Represents the prolonged tail of the game where only highly dedicated users remain. Any level l not satisfying the conditions for the previous three phases is classified as Late Game.

2.3 Algorithmic Representation

The following pseudocode details the exact procedural logic used in the data pipeline to tag these phases.

Algorithm 1 Classify Levelset Phases

Require: DataFrame D for a single Levelset
Ensure: DataFrame D with assigned 'Phase' column

Filter D to include only levels ≤ 250
if length(D) < 3 **then**
    return Null ▷ Insufficient data
end if
$l_{max} \leftarrow \max(D.\text{level\_no})$
D.phase $\leftarrow$ Empty Array
for each row i in D **do**
    $l \leftarrow D.\text{level\_no}[i]$
    $r \leftarrow D.\text{retention\_pct}[i]$
    if $l \leq 3$ **or** $r \geq 85$ **then**
        $D.\text{phase}[i] \leftarrow$ 'Onboarding'
    end if
end for
$drop\_indices \leftarrow$ Find all indices where $D.\text{retention\_pct} < 60$
if $drop\_indices$ is not empty **then**
    $first\_drop \leftarrow drop\_indices[0]$
else
    $first\_drop \leftarrow \text{length}(D)$
end if
for each row i in D **do**
    if $D.\text{phase}[i]$ is empty **and** $i < first\_drop$ **then**
        $D.\text{phase}[i] \leftarrow$ 'Early Game'
    end if
end for
for each row i in D **do**
    $l \leftarrow D.\text{level\_no}[i]$
    $r \leftarrow D.\text{retention\_pct}[i]$
    if $D.\text{phase}[i]$ is empty **and** ($r \geq 30$ **or** $l \leq 0.3 \times l_{max}$) **then**
        $D.\text{phase}[i] \leftarrow$ 'Mid Game'
    end if
end for
for each row i in D **do**
    if $D.\text{phase}[i]$ is empty **then**
        $D.\text{phase}[i] \leftarrow$ 'Late Game'
    end if
end for
return D

3 Level Difficulty Clustering

To categorize the inherent complexity of each level without relying on dynamic player behavior (such as hint usage or churn), an unsupervised learning approach was applied exclusively to the game's static design features. This analysis was similarly restricted to the first 250 levels of each levelset.

3.1 Feature Extraction

For each level, the raw structural data is parsed to extract key design metrics. Let W represent the set of all words in a given level. For each word $w \in W$, let C_w denote the set of spatial coordinates (x, y) occupied by the letters of w on the grid.

We extract the following feature vector $\vec{f}$ for each level:

- **Word Count ($|W|$):** The total number of words on the board.
- **Board Dimensions (X, Y):** The width (`dimx`) and height (`dimy`) of the grid.
- **Board Count (B):** The number of distinct puzzle boards within the level.
- **Word Lengths (L):** We calculate the average (μ_L), maximum ($max(L)$), and minimum ($min(L)$) lengths of the words in W.
- **Seed Word Length (S):** The character length of the base word provided to the user.
- **Intersections (I):** The total number of grid cells where two or more words overlap. Mathematically, this is the cardinality of the set of coordinates that exist in multiple C_w:

$$I = \left| \left\{ (x, y) \;\middle|\; \sum_{w \in W} \mathbf{1}_{\{(x,y) \in C_w\}} \geq 2 \right\} \right| \tag{5}$$

3.2 Normalization and Clustering

To ensure all structural features contribute equally to the distance metric, the feature matrix F is standardized using Z-score normalization. For each feature column j, the scaled value $z_{i,j}$ is computed as:

$$z_{i,j} = \frac{f_{i,j} - \mu_j}{\sigma_j} \tag{6}$$

We apply the K-Means clustering algorithm to partition the scaled levels into $k = 5$ distinct clusters. The algorithm minimizes the within-cluster sum of squares (WCSS):

$$\arg\min_{S} \sum_{c=1}^{5} \sum_{\vec{z} \in S_c} \|\vec{z} - \vec{\mu}_c\|^2 \tag{7}$$

where S_c represents the set of levels assigned to cluster c, and $\vec{\mu}_c$ is the centroid of cluster c.

3.3 Difficulty Ranking

Because K-Means assigns cluster labels arbitrarily, we must map these clusters to an ordinal difficulty scale (1 = Easiest, 5 = Hardest).

A composite difficulty score is calculated for each cluster centroid based on primary complexity drivers: board count, word count, intersections, and grid dimensions. Let $\bar{z}_{c,j}$ denote the value of feature j for the centroid of cluster c. The difficulty score D_c is defined as:

$$D_c = \frac{1}{5} \left(\bar{z}_{c,B} + \bar{z}_{c,|W|} + \bar{z}_{c,I} + \bar{z}_{c,X} + \bar{z}_{c,Y} \right) \tag{8}$$

The clusters are then sorted by D_c in ascending order, and the ranked indices 1 through 5 are mapped back to the levels as their final categorical difficulty rating.

3.4 Algorithmic Representation

Algorithm 2 Extract Features and Assign Difficulty

Require: Dataset of Levels D_{raw} (filtered to level ≤ 250)
Ensure: Dataset D_{final} with assigned 'Difficulty' (1 to 5)

$F \leftarrow$ Empty Matrix
for each level l in D_{raw} **do**
    Extract $B, X, Y, |W|, \mu_L, max(L), S$
    Compute Intersections I by mapping all word coordinates
    Append feature vector $\vec{f_l}$ to F
end for
$F_{scaled} \leftarrow$ Standardize(F) ▷ Z-score normalization
$C \leftarrow$ KMeansFit($F_{scaled}, k = 5$)
Centroids $M \leftarrow C$.cluster_centers
$Scores \leftarrow$ Empty Array
for each centroid $\vec{\mu}_c \in M$ **do**
    $Score_c \leftarrow$ Mean($\vec{\mu}_c[B], \vec{\mu}_c[|W|], \vec{\mu}_c[I], \vec{\mu}_c[X], \vec{\mu}_c[Y]$)
    Append $Score_c$ to $Scores$
end for
$RankedClusters \leftarrow$ Argsort($Scores$) ▷ Ascending order
$DifficultyMap \leftarrow$ Create mapping from c to Rank $(1 \dots 5)$
for each level l **do**
    $l.Difficulty \leftarrow DifficultyMap[l.Cluster]$
end for
return D_{raw} with appended feature columns and Difficulty

4 Phase-Based Difficulty and Retention Analysis

To understand how the pacing of difficulty impacts player churn, the previously classified game phases were cross-referenced with the K-Means difficulty scores (ranked 1 to 5). A comparative statistical analysis was performed between the highest-retention levelset (ID 1367) and the lowest-retention levelset (ID 1661).

4.1 Methodology

For each distinct phase $P \in \{\text{Onboarding}, \text{Early Game}, \text{Mid Game}, \text{Late Game}\}$, let $D_{best,P}$ and $D_{worst,P}$ represent the distributions of difficulty scores for the highest and lowest retention levelsets, respectively.

To evaluate whether a statistically significant difference exists between the difficulty distributions of the two levelsets, an independent two-sample Welch's t-test was conducted. This test does not assume equal variances and is well-suited for unequal sample sizes. The test statistic t is defined as:

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{s_1^2}{N_1} + \frac{s_2^2}{N_2}}} \tag{9}$$

where $\bar{X}$ is the sample mean, s^2 is the sample variance, and N is the sample size.

Additionally, a non-parametric Mann-Whitney U test was applied to account for the ordinal nature of the difficulty scores (1 through 5), providing a robust secondary validation.

4.2 Results and Statistical Summary

The comparative results across all four phases are summarized in Table 1. During the Onboarding and Early Game phases, both levelsets exhibited a mean difficulty of exactly 1.0 (variance of 0), reflecting identical, highly-controlled introductory experiences.

4.3 Discussion on Significant Findings

A notable and statistically significant divergence occurs exclusively during the **Mid Game** phase.

Table 1: Statistical Comparison: Best (1367) vs Worst (1661) Retention

Phase	**Best Mean**	**Worst Mean**	**Best** n	**Worst** n	**t-test** p	**M-W** p
Onboarding	1.0000	1.0000	3	3	NaN	1.0000
Early Game	1.0000	1.0000	8	4	NaN	1.0000
Mid Game	**2.3281**	**1.9118**	64	68	**0.0049**	0.0631
Late Game	3.7200	3.6686	175	175	0.6237	0.8046

The high-retention levelset introduces a higher average difficulty ($\mu = 2.33$) compared to the low-retention levelset ($\mu = 1.91$). Welch's t-test confirms this difference is highly significant ($p = 0.0049 < \alpha = 0.05$).

This indicates that **higher player retention is associated with a steeper difficulty ramp during the Mid Game.** Rather than causing churn, introducing moderately higher complexity (specifically via design features like board count, intersections, and word count) during the levels following the Early Game appears to engage players more effectively, likely by providing adequate mental stimulation before the severe difficulty plateau of the Late Game.

5 Mid Game Difficulty Progression Analysis

Following the discovery of a statistically significant divergence in mean difficulty during the Mid Game phase, a granular sequential analysis was conducted. Rather than comparing absolute level numbers—which may misalign due to differing onboarding lengths—the levels were normalized by their progression sequence within the phase itself.

5.1 Sequential Alignment Methodology

For each levelset, the Mid Game phase was isolated and indexed chronologically. Let L_P denote the ordered set of levels classified as Mid Game for a given levelset. For each level $l_i \in L_P$, we assign a sequence index $s_i \in \{1, 2, \ldots, N\}$, where N is the total number of Mid Game levels in that levelset.

This relative alignment ensures that the 1^{st} level of the high-retention Mid Game is directly compared to the 1^{st} level of the low-retention Mid Game, preserving the player's chronological experience of the difficulty curve.

5.2 Progression Statistics and Variance

The difficulty scores assigned via the prior K-Means clustering (1=Easiest, 5=Hardest) were mapped across this normalized sequence. The summary statistics for the highest retention (ID 1367) and lowest retention (ID 1661) levelsets are presented in Table 2.

Table 2: Mid Game Difficulty Summary (Ranked 1-5)

Levelset Profile	**Levelset ID**	**Mean** (μ)	**Standard Deviation** (σ)	**Min**	**Max**
Best Retention	1367	2.33	**1.05**	1	5
Worst Retention	1661	1.91	**0.48**	1	3

5.3 Discussion: Pacing Volatility and the Standard Deviation

While the higher mean difficulty ($\mu = 2.33$ vs $\mu = 1.91$) is notable, the most critical differentiator in the Mid Game pacing is the **Standard Deviation**.

The high-retention levelset exhibits more than double the variance ($\sigma = 1.05$) of the low-retention levelset ($\sigma = 0.48$). An analysis of the sequential progression reveals that the low-retention levelset is highly monotonous, fluctuating only slightly between difficulties 1, 2, and occasionally 3.

Conversely, the high-retention levelset utilizes high pacing volatility. It introduces significant difficulty spikes (reaching levels 4 and 5) that are immediately followed by cognitive relief periods (dropping back to difficulties 1 and 2). This wave-like difficulty distribution—rather than a flat, monotonous curve—appears essential for maintaining player flow and preventing churn during the game's core loop.

6 High-Resolution Validation: 10-Point Difficulty Scale

To ensure that the observed differences in pacing and volatility were not artifacts of broad categorization or misclassification within the 5-point system, the unsupervised clustering pipeline was rerun with increased granularity. The K-Means algorithm was configured to partition the data into $k = 10$ clusters, effectively creating a 1-to-10 scale of difficulty based on the same fundamental design features (board count, dimensions, word count, and intersections).

6.1 Mid Game Re-Evaluation

The chronologically aligned Mid Game sequences for both the highest retention (ID 1367) and lowest retention (ID 1661) levelsets were mapped to this new 10-point scale. The summary statistics are detailed in Table 3.

Table 3: Mid Game Difficulty Summary (Ranked 1-10)

Levelset Profile	Levelset ID	Mean (μ)	Standard Deviation (σ)	Min	Max
Best Retention	1367	3.27	**2.20**	1	8
Worst Retention	1661	2.16	**1.18**	1	5

6.2 Validation of Pacing Volatility

The transition to a higher-resolution scale strongly corroborates the initial findings, proving that the pacing differences are systemic rather than categorical artifacts:

- **Sustained Mean Difference:** The high-retention levelset continues to demand a higher average cognitive load ($\mu = 3.27$) compared to the low-retention levelset ($\mu = 2.16$).
- **Pronounced Variance:** Crucially, the standard deviation metric confirms the pacing strategy. The high-retention levelset exhibits nearly double the volatility ($\sigma = 2.20$) of the underperforming levelset ($\sigma = 1.18$).

6.3 Sequential Pacing Observations

An inspection of the sequential progression data reveals the distinct architectural philosophies between the two levelsets. The underperforming levelset (1661) suffers from severe difficulty stagnation; for over 38 consecutive levels within the core Mid Game, the difficulty almost exclusively toggles between scores of 1 and 2, with only a singular jump to 5.

Conversely, the optimal levelset (1367) utilizes a highly deliberate "spike and cooldown" wave mechanic. It establishes a baseline of low-difficulty levels (1s and 2s) but consistently punctuates them with severe difficulty spikes (reaching scores of 5 and 8). Importantly, these extreme spikes are almost always followed immediately by a return to a difficulty of 1 or 2, providing necessary cognitive relief. This highly elastic pacing curve appears to be a primary driver of sustained player engagement.

7 Comprehensive Multi-Levelset Mid Game Analysis

To robustly validate the hypothesis that pacing volatility and higher average difficulty drive player engagement, the 10-point Mid Game analysis was expanded to encompass all four available levelsets (IDs 1367, 1421, 1661, and 1671). This broader dataset allows for a definitive ranking and correlation analysis between structural difficulty and actual user retention.

7.1 Overall Mid Game Performance Metrics

Table 4 presents the consolidated summary statistics for the Mid Game phase across all levelsets, ordered by average retention percentage.

An inverse relationship between consistency and retention is immediately apparent. The best-performing levelsets exhibit the highest average difficulty (**Diff** μ) and the highest volatility (**Diff** σ), while the worst-performing levelsets are mathematically the easiest and most consistent.

Table 4: Mid Game Metrics Summary (Ranked by Average Retention)

Levelset	N Levels	Diff μ	Diff σ	Ret μ	Ret σ	Pearson r
1367 (Best)	64	3.266	2.198	**30.55%**	9.80%	-0.356
1421	67	3.119	2.012	24.35%	11.30%	-0.471
1671	68	1.912	0.748	20.48%	13.16%	-0.512
1661 (Worst)	68	2.162	1.180	19.30%	13.10%	-0.396

7.2 Statistical Validation

To quantify these observations, Welch's t-tests and Mann-Whitney U tests were applied, comparing the top-performing levelset (1367) against the remaining cohorts.

- **1367 vs. Underperformers (1661 & 1671):** The highest-retention levelset is significantly harder than both underperforming levelsets ($p < 0.001$). Correspondingly, its retention is also significantly higher ($p < 0.001$).
- **1367 vs. 1421:** There is no statistically significant difference in difficulty between the top two levelsets ($p = 0.6923$). However, 1367 still achieves significantly higher retention ($p = 0.0010$), suggesting that while high volatility is necessary, the exact structural placement of those difficulty spikes plays an optimal role in levelset 1367.

7.3 Architectural Pacing and the "Spike" Mechanic

To map how this variance manifests chronologically, we analyzed the sequential progression of each Mid Game, defining a "Difficulty Spike" as an abrupt sequential increase of $\Delta\text{Difficulty} \geq 2$.

1. **High-Retention Cohort (1367 & 1421):** Both levelsets employ a highly aggressive "spike and cooldown" mechanic. Levelset 1367 triggers 13 distinct difficulty spikes (with jumps as high as +7), while 1421 triggers 15 spikes (jumping up to +8). This forces high cognitive load followed by immediate relief, establishing a deep state of psychological flow.
2. **Low-Retention Cohort (1661 & 1671):** These levelsets suffer from severe pacing stagnation. Levelset 1661 contains only 6 spikes (largely clustered at the very end of the phase), while 1671 is entirely flat, containing only 1 spike across its 68 Mid Game levels.

7.4 Conclusion

The data categorically refutes the assumption that "easier and smoother" gameplay retains users. Instead, **cognitive volatility is a primary driver of retention**. Players require a baseline of manageable levels heavily punctuated by severe, abrupt challenges. Levelsets that fail to introduce these regular difficulty spikes (low standard deviation) result in player boredom and significantly higher churn rates.

8 Impact of Vocabulary Repetition on Churn

Following the structural difficulty analysis, we investigated the semantic design of the levels—specifically, whether repeating words within a player's recent memory influences their likelihood to churn. We hypothesize that encountering a previously solved word provides a "cognitive anchor" (or familiarity relief) that mitigates player frustration, especially during mechanically complex levels.

8.1 Macro-Level Analysis: Global Efficacy

To establish a baseline for this phenomenon, an aggregate analysis was conducted across the entire ∼2,000 level lifespan of all four levelsets. A static short-term memory window of 10 levels was assumed.

For each level i, churn is defined as the percentage of the player base lost since the preceding level:

$$\text{Churn}_i = \left(\frac{U_{i-1} - U_i}{U_{i-1}} \right) \times 100 \tag{10}$$

A level was flagged as containing a "Repeated Word" if at least one word in its solution set appeared in any of the preceding 10 levels. Independent Welch's t-tests were conducted for each levelset, summarized in Table 5.

Table 5: Global Churn Impact of Repeated Words (10-Level History Window)

Levelset	N **(Rep)**	N **(Not Rep)**	μ **Churn (Rep)**	μ **Churn (Not Rep)**	Δ **Churn**	**p-value**
1367 (Best)	1,274	725	0.16%	0.39%	-0.24%	< 0.001
1421	1,525	474	0.19%	0.60%	-0.41%	< 0.001
1661 (Worst)	1,274	725	0.24%	0.69%	-0.45%	< 0.001
1671	1,165	834	0.19%	0.69%	-0.50%	< 0.001
Aggregate	**5,238**	**2,758**	**0.20%**	**0.60%**	**-0.40%**	$\mathbf{< 0.001}$

Across all levelsets combined, the baseline churn for a level with entirely novel vocabulary is 0.60%. When a single repeated word is introduced, average churn plummets to 0.20%. **Simply repeating a word from the recent past reduces level-to-level churn by exactly 66%.**

serving as a blueprint for optimal level design. The Odds Ratios (OR), defined as e^{β}, represent the multiplicative change in the odds of a user churning. An OR < 1.0 indicates a reduction in churn.

Table 6: GEE Churn Model Odds Ratios (Mid and Late Game)

Predictor	**Mid Game OR**	**Mid Game** p**-value**	**Late Game OR**	**Late Game** p**-value**
`level_no`	0.9816	< 0.001*	0.9956	< 0.001*
`difficulty_10`	1.0175	0.323	1.0143	0.125
`diff_masd`	**0.5599**	< 0.001*****	**0.6267**	< 0.001*****
`repeated_word`	1.0387	0.659	**0.8326**	**0.008***
`min_zipf`	0.8038	< 0.001*	1.0046	0.915

8.2 Discussion of Key Findings

The multivariate model confirms our core hypotheses while revealing how the importance of certain mechanics shifts as the player matures.

1. **The Supremacy of Volatility (MASD):** The most powerful controllable metric in the game is pacing volatility. In both the Mid Game (OR = 0.5599) and Late Game (OR = 0.6267), higher `diff_masd` drastically reduces churn. A 1-point increase in volatility reduces the odds of churning by roughly 37% to 44%, completely overpowering the negligible impact of absolute difficulty (`difficulty_10`), which showed no statistical significance. This proves definitively that *how* difficulty changes is vastly more important than *how hard* the game is.

2. **Phase-Dependent Semantic Anchoring:** The protective effect of repeating vocabulary (`repeated_word`) becomes mathematically significant in the Late Game (OR = 0.8326). As structural complexity reaches its zenith, providing players with familiar words reduces churn odds by nearly 17%.

8.3 Conclusion

To maximize user lifetime value, designers should disregard absolute difficulty caps. Instead, the focus must be entirely on the derivative of difficulty: maximizing sequential variance (high MASD) to prevent stagnation, while strategically deploying repeated vocabulary to buffer cognitive overload during the harshest spikes.